

CMSE 801 - Day 18

Autoregression

Nov 9, 2022

General Course Announcements

- Homework #4 is due Monday Nov 14 at 11:59 p.m.
- Project's deadline Sunday Dec 4 at 11:59 p.m.
- Homework #5 will be released today and is due Wednesday Nov 30 at 11:59 p.m.
- Homework #5 is extra credit, you can gain up to 5% of the final grade

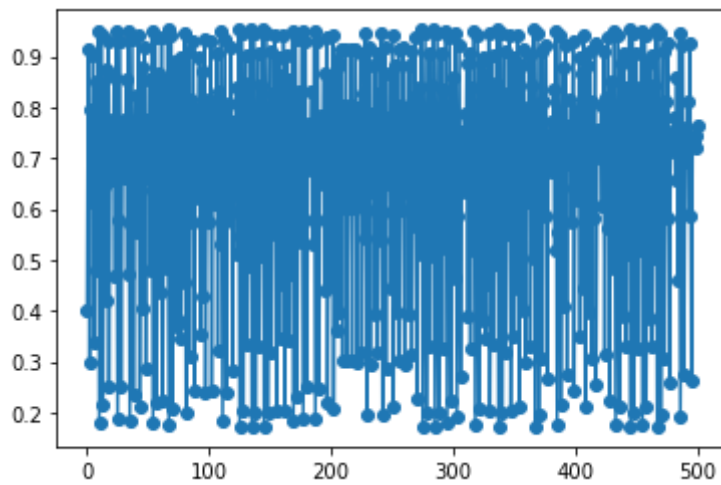
Questions/comments from Day 18 PCA

- Could you explain the last question? - the relationship between actual value and predicted value from polynomials of different degrees
- If data has auto correlation, does it mean it has some kind of correlation with time?
- Is there a more direct way to predict a lot more points with autoregression other than using a loop?
- I got a value error in the last part, is that possible to you to go through the last part of the Pre-Class!!?
- So if we have data with autocorrelation, we can predict only one data point each time?
- I'm curious as to why I can just use 'plot_lag' and I do not have to type 'pd.' before it.?
- when we compute a mask then the result is a copy of the original data but with false and true instead of the values?
- Not right now! I am enjoying the process of learning these concepts even though sometimes I bang my head against the wall.

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
test_input = np.array([0.4,0.9144,0.298219,0.797373,0.615578,0.901605,0.337999,0
```

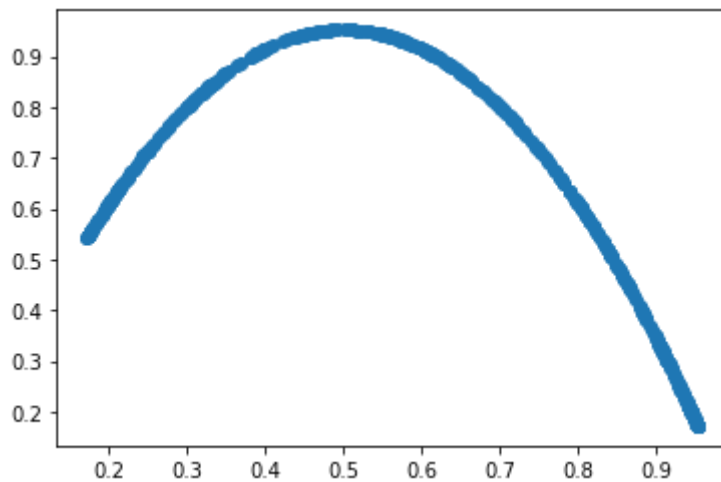
```
In [2]: plt.plot(test_input,marker='o')
```

```
Out[2]: [<matplotlib.lines.Line2D at 0x7f976ee18730>]
```



```
In [3]: plt.scatter(test_input[0:len(test_input)-1],test_input[1:len(test_input)])
```

```
Out[3]: <matplotlib.collections.PathCollection at 0x7f976f21af70>
```

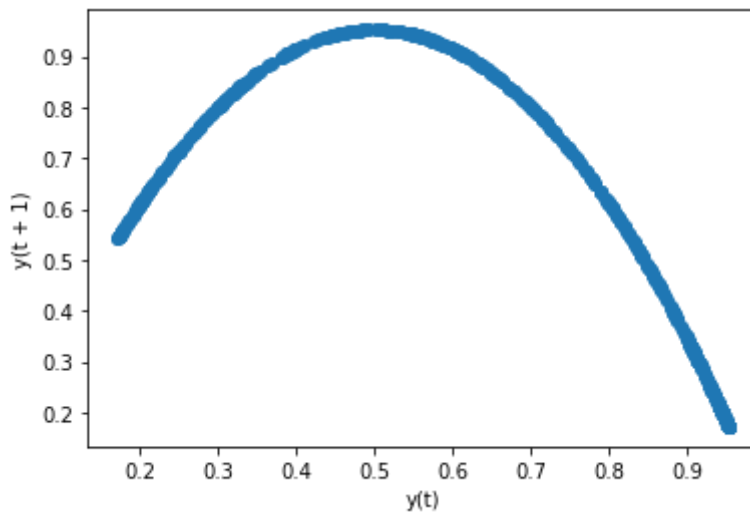


```
In [4]: print(pd.Series(test_input))
```

```
0      0.400000
1      0.914400
2      0.298219
3      0.797373
4      0.615578
...
496    0.740189
497    0.732697
498    0.746196
499    0.721567
500    0.765460
Length: 501, dtype: float64
```

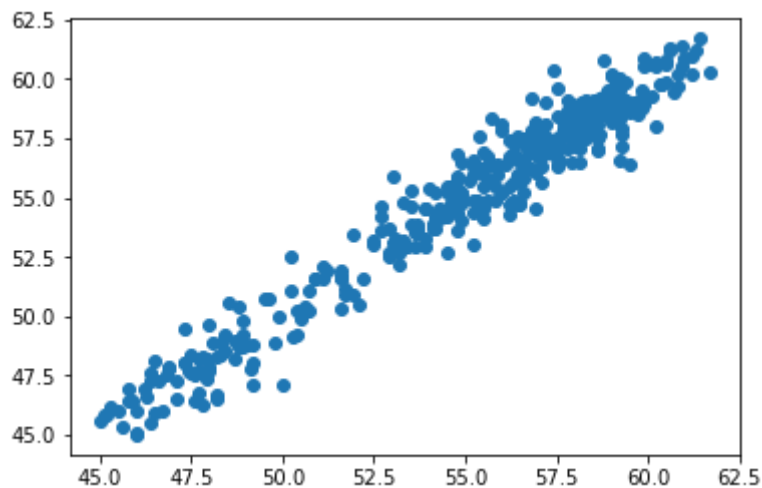
```
In [5]: from pandas.plotting import lag_plot
lag_plot(pd.Series(test_input))
```

```
Out[5]: <AxesSubplot:xlabel='y(t)', ylabel='y(t + 1)'>
```



```
In [6]: trump = pd.read_csv("trump_ratings_first500.csv")
ipsos = trump['pollster'] == 'Ipsos'
trump = trump[ipsos]
```

```
In [7]: plt.scatter(trump['disapprove'].iloc[0:399], trump['disapprove'].iloc[1:400])
pfit = np.polyfit(trump['disapprove'].iloc[0:399], trump['disapprove'].iloc[1:400])
autoreg = np.polyld(pfit)
```



```
In [8]: starting_date = trump['datenumber'].iloc[400]
starting_value = trump['disapprove'].iloc[400]
num_predictions = 100
forecast_dates = []
forecast_values = []
for i in range(num_predictions):
    date = starting_date + 1
    new_value = autoreg(starting_value) # pfit[0] * starting_value + pfit[1]
    forecast_dates.append(date)
    forecast_values.append(new_value)
    starting_date = date
    starting_value = new_value
```

```
In [9]: plt.figure(2, figsize=(16,4))
starting_date = trump['datenumber'].iloc[400]
plt.plot([starting_date, starting_date], [45, 60], color = 'grey')
```

```
plt.plot_date(trump['datenumber'],trump['disapprove'], marker = '+')  
plt.plot_date(forecast_dates, forecast_values)
```

Out[9]: [

